FLATNESS IS STABLE UNDER BASE CHANGE OF FORMAL ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof delegates to the continuous-family base-change lemma and transfers the outcome through the property-defines-property lemmas of the chapter

Editorial note. The label lemma-base-change-flat names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the two base-change comparison remarks of Formal Spaces for Noetherian formal algebraic spaces (formal-spaces-remark-base-change-variant-variant-Noetherian and formal-spaces-remark-base-change-Noetherian)
- base-change stability of flatness for continuous families and its locally-finite-type companion (lemma-base-change-flat-continuous of the same chapter, parts (1) and (2))
- flatness and the two finite-type notions as properties of morphisms read through etale covers (lemma-flat-morphisms, lemma-finite-type-red-morphisms, lemma-finite-type-morphisms of the same chapter)

Source reference. Original remark 0GBF: sources/stacks-chow/tags.txt, source lines 20370–20370. Prerequisite pointer invoked at proof line(s) 3116; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 0GBG: sources/stacks-chow/tags.txt, source lines 20371–20371. Prerequisite pointer invoked at proof line(s) 3108; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GCA: sources/stacks-restricted/pinned-original-source.tex, source lines 2951–2951. Prerequisite pointer invoked at proof line(s) 3109; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GC0: sources/stacks-restricted/pinned-original-source.tex, source lines 2556–2556. Prerequisite pointer invoked at proof line(s) 3119; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GC7: sources/stacks-restricted/pinned-original-source.tex, source lines 2795–2795. Prerequisite pointer invoked at proof line(s) 3111; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GCD: sources/stacks-restricted/pinned-original-source.tex, source lines 3043–3043. Prerequisite pointer invoked at proof line(s) 3110; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ and $g : Z \to Y$ be morphisms of locally Noetherian formal algebraic spaces over S.*

(1) *If f is flat and $g_{red} : Z_{red} \to Y_{red}$ is locally of finite type, then the base change $X \times_Y Z \to Z$ is flat.*
(2) *If f is flat and locally of finite type, then the base change $X \times_Y Z \to Z$ is flat and locally of finite type.*

Proof. Part (1) follows from a combination of Formal Spaces, Remark 0GBG, Lemma 0GCA part (1), Lemma 0GCD, and Lemma 0GC7.

Part (2) follows from a combination of Formal Spaces, Remark 0GBF, Lemma 0GCA part (2), Lemma 0GCD, and Lemma 0GC0. □